

Judith Hoeller

Postdoctoral Associate

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EDUCATION

Ph.D. in physics, Yale University, New Haven, USA (Sep. 2014-Dec. 2020)

M.S. in physics, ETH Zurich, Switzerland (Sep. 2012-April 2014)

B.S. in mathematics, ETH Zurich, Switzerland (Sep. 2012-April 2014)

B.S. in physics, ETH Zurich, Switzerland (Sep. 2009-Sep. 2012)

RESEARCH EXPERIENCE

Postdoctoral associate in neuroscience (Sep. 2020-present)

HHMI Janelia Research Campus, Ashburn, USA

with Dr. Sandro Romani

- developed a group-theoretic framework to analyze neuronal population responses
- designed calcium imaging and behavioral experiments in mice
- analyzed high-resolution electron-microscopy images of the whole visual system of the fruit fly in collaboration with a diverse team of scientists

Graduate student in physics (Sep. 2015-Aug. 2020)

Yale University, New Haven, USA

with Prof. Nicholas Read

Thesis: *Topological Quantization of Berry Phases in Quantum and Classical Systems*

- proved rigorous results on topological insulators using group theory
- studied phase transitions in short-range spin glasses using field theory
- collaborated with various experimental groups on a multitude of interacting systems

Semester project in physics (Jan. 2015-June 2015)

Yale University, New Haven, USA

with Prof. Eric Brown

Project: *Universal Fluctuations in Rayleigh-Bénard Convection*

Internship in R&D at OptoTune AG, Switzerland (May 2014-July 2014)

- programmed a spectrometer to measure the thicknesses of optical lenses
- simulated the impact of temperature gradients on optical properties

Master project in physics (Sep. 2013-April 2014)

ETH Zurich, Switzerland

with Prof. Atac Imamoglu

Project: *Lévy Flights in Coherence Times of Solid State Qubits*

Semester project in physics (Jan. 2013-Aug. 2013)

ETH Zurich, Switzerland

with Prof. Atac Imamoglu

Project: *Periodic Trapping in a Single Quantum Dot*

Undergraduate project in mathematics (Sep. 2013-Dec. 2013)

ETH Zurich, Switzerland

with Prof. Damien Calaque

Project: *Factorization Algebras in Quantum Mechanics*

Advanced lab in physics (Jan. 2012-July 2012)

ETH Zurich, Switzerland

- fabricated a Hall bar and measured its Quantum Hall effect
- measured the dispersion of lead using a synchrotron (Swiss Light Source)
- entangled photons to test Bell's inequality on quantum mechanics

AWARDS

As a postdoc at HHMI Janelia, I am not eligible to apply for external funding.

Travel award for Excellence in Graduate Research (\$500, 2016)

from the Forum on Graduate Student Affairs of the American Physical Society

Yale Leigh Page Prize for incoming graduate students (\$2000, 2014)

in recognition of a fine academic record

TEACHING & MENTORING EXPERIENCE

Mentor of undergraduate (Janelia-Meyerhoff program, 10 weeks in 2023)

HHMI Janelia Research Campus, Ashburn, USA

- designed and guided a summer research project
- taught programming and how to analyze data

Teaching assistant in physics (6 semesters in 2014-2019)

Yale University, New Haven, USA

- held discussion sessions for introductory life sciences and quantum mechanics
- supervised lab experiments in electronics and optics
- prepared incoming physics graduate students for their qualifying exams

Teaching assistant in mathematics (3 semesters in 2012-2013)

ETH Zurich, Switzerland

- held discussion sessions on linear algebra and probability theory

SUMMER SCHOOLS (participated)

Scientists Teaching Science, virtual (40 hours in 2024)

Methods in Computational Neuroscience, Woods Hole, USA (2021)

Brains, Minds & Machines, virtual (2020)

Topological Phases of Quantum Matter, Boulder, USA (2016)

MEETINGS & WORKSHOPS

Analysis and Modeling of Connectomes, Ashburn, USA (2025)

Poster: *Visual Information Flow and Pathway Segregation in the Fruit Fly Visual System*

German Neuroscience Society, Goettingen, Germany (2025)

Poster: *Bridging Tuning and Invariance with Equivariant Neuronal Representations*

Society for Industrial and Applied Mathematics, Portland, USA (2024)

Talk: *Equivariant Neuronal Representations*

Computational and Systems Neuroscience, Lisbon, Portugal (2024)

Poster: *How populations of neurons in visual cortex encode perspective*

Society for Neuroscience, Washington D.C., USA (2023)

Poster: *How populations of neurons in visual cortex encode perspective*

Invertebrate Vision, Bäckaskog Castle, Sweden (2023)

Society for Neuroscience, San Diego, USA (2022)

Poster: *Slant coding in the mouse visual cortex*

Vision Sciences Society, St. Pete Beach, USA (2022)

Talk: *Compositional & texture-independent 3D orientation coding in visual cortex*

Computational and Systems Neuroscience, Lisbon, Portugal (2022)

Computational and Systems Neuroscience, virtual (2021)

American Physical Society, Boston, USA (2019)

Talk: *Non-Hermitian adiabatic transport in the space of exceptional points*

American Physical Society, Los Angeles, USA (2018)

Talk: *Do more symmetries lead to more topological phases?*

Topological States and Phase Transitions in Strongly Correlated Systems, Chinese Academy of Sciences, Beijing, China (2017)

Strongly Correlated Topological Phases of Matter, Simons Center for Geometry and Physics, Stony Brook, USA (2017)

American Physical Society, New Orleans, USA (2017)

Talk: *Topological Bloch oscillations*

American Physical Society, Baltimore, USA (2016)

Talk: *Real-time monitoring of Lévy flights in a single quantum system*

Topological Quantum Matter: Progress and Applications, Aspen Center for Physics, USA (2016)

SEMINARS

Washington University in St. Louis, Center for Theoretical and Computational Neuroscience (2025)

Insights into the Neural Code of Vision from Mice, Flies, and AI

Max Planck Institute Martinsried, Biological Intelligence (2025)

Insights into the Neural Code of Vision from Mice, Flies, and AI

Wuerzburg University, Wuerzburg Insect Research (2025)

Insights into the Neural Code of Vision from Mice, Flies, and AI

New York University, Biophysics (2025)

Insights into the Neural Code of Vision from Mice, Flies, and AI

Janelia Research Campus, Computation and Theory (2020)

How to topologically oscillate and braid

Washington University in St. Louis, Condensed Matter, Materials & Biological Physics (2020)
Topological Quantization of Berry Phases in the Space of Exceptional Points

University of Illinois Urbana-Champaign, Institute for Condensed Matter (2019)
Topological Berry phases in Quantum and Classical systems

Harvard University, Institute for Theoretical Atomic, Molecular and Optical Physics (2019)
Topological Quantization of Berry Phases in the Space of Exceptional Points

Yale University, Quantum Institute (2019)
Topological Quantization of Berry Phases in the Space of Exceptional Points

Yale University, Wright Laboratory (2019)
Critical coupling of weakly-damped harmonic oscillators

Ludwig Maximilians University of Munich, group of Prof. I. Bloch (2018)
Topological Bloch oscillations

Yale University, groups in Atomic, Molecular and Optics (2018)
Topological Bloch oscillations

Chinese Academy of Sciences Beijing, CAS Key Lab of Condensed Matter Theory & Materials Computation (2017)
Topological Bloch oscillations

Ludwig Maximilians University of Munich, group of Prof. I. Bloch (2016)
Topological invariant fractional Bloch oscillations

ACADEMIC SERVICE

co-organized the Junior Scientist Workshop on Theoretical Neuroscience at Janelia (2025)

peer-reviewed for Nature (2023) and Physical Review (2018-2020)

co-organized the Gotham-Metro conference in soft and hard condensed matter at New York University (2019)

co-organized the symposium *Field Theory in Condensed Matter* at Yale University (2019)

organized the condensed matter journal club at Yale University (2018-2019)

organized the graduate student lunch seminar at Yale University (2016-2017)